CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the textbook exercises: 20 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 7.1

Q1 Find the distance between each pair of points: (i) $(2, 3)$, $(4, 1)$ (ii) $(-5, 7)$, $(-1, 3)$ (iii) (a, b), $(-a, -b)$.

SOLUTION

Use the distance formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

(i) $\sqrt{(4-2)^2 + (1-3)^2} = \sqrt{4+4} = 2\sqrt{2}$

(ii) $\sqrt{(-1+5)^2 + (3-7)^2} = \sqrt{16+16} = 4\sqrt{2}$

(iii) $\sqrt{(-2a)^2 + (-2b)^2} = \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2 + b^2}$

Answer: (i) $2\sqrt{2}$ (ii) $4\sqrt{2}$ (iii) $2\sqrt{a^2 + b^2}$ units

Q2 Find the distance between $(0, 0)$ and $(36, 15)$. Hence find the distance between the two towns of Section 7.2 of the textbook, where town B is 36 km east and 15 km north of town A.

SOLUTION

Distance $= \sqrt{36^2 + 15^2} = \sqrt{1296 + 225} = \sqrt{1521} = 39$.

Taking town A as the origin, town B is the point $(36, 15)$, so the towns are 39 km apart.

Answer: 39 units, so the towns are 39 km apart

Q3 Are the points $(1, 5)$, $(2, 3)$ and $(-2, -11)$ collinear?

SOLUTION

Call them A, B and C. $AB = \sqrt{1+4} = \sqrt{5} \approx 2.24$, $BC = \sqrt{16+196} = \sqrt{212} \approx 14.56$ and $AC = \sqrt{9+256} = \sqrt{265} \approx 16.28$.

The two shorter distances add up to about 16.80, which is not equal to the longest distance 16.28.

Answer: no, the points are not collinear

Q4 Are $(5, -2)$, $(6, 4)$ and $(7, -2)$ the vertices of an isosceles triangle?

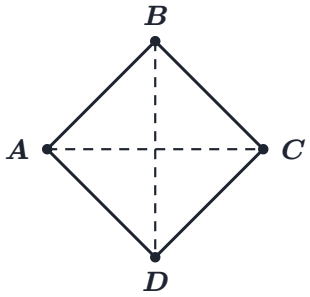
SOLUTION

Call them A, B and C. $AB = \sqrt{1+36} = \sqrt{37}$, $BC = \sqrt{1+36} = \sqrt{37}$ and $AC = \sqrt{4+0} = 2$.

$AB = BC$, and the points are not collinear since $AB + AC \neq BC$.

Answer: yes, the triangle is isosceles with $AB = BC$

Q5 Four friends sit at the points $A(3,4)$, $B(6,7)$, $C(9,4)$ and $D(6,1)$ of a classroom grid (read from the textbook figure). One student says $ABCD$ is a square and another disagrees. Who is correct?



A(3, 4), B(6, 7), C(9, 4), D(6, 1)

SOLUTION

Sides: $AB = \sqrt{3^2+3^2} = \sqrt{18}$, $BC = \sqrt{3^2+3^2} = \sqrt{18}$, $CD = \sqrt{18}$ and $DA = \sqrt{18}$. All four sides are equal.

Diagonals: $AC = \sqrt{6^2+0^2} = 6$ and $BD = \sqrt{0^2+6^2} = 6$. The diagonals are equal.

A quadrilateral with four equal sides and equal diagonals is a square.

Answer: $ABCD$ is a square, so the student who said so is correct

Q6 Name the type of quadrilateral formed, if any, by these points, with reasons: (i) $(-1,-2)$, $(1,0)$, $(-1,2)$, $(-3,0)$ (ii) $(-3,5)$, $(3,1)$, $(0,3)$, $(-1,-4)$ (iii) $(4,5)$, $(7,6)$, $(4,3)$, $(1,2)$.

SOLUTION

Call the points A, B, C, D in order.

(i) $AB = BC = CD = DA = \sqrt{4+4} = \sqrt{8}$. Diagonals: $AC = 4$ and $BD = 4$. All sides are equal and the diagonals are equal, so it is a square.

(ii) $AB = \sqrt{36+16} = 2\sqrt{13}$, $BC = \sqrt{9+4} = \sqrt{13}$ and $AC = \sqrt{9+4} = \sqrt{13}$. Since $AC + CB = AB$, the points A, C, B lie on one line, so no quadrilateral is formed.

(iii) $AB = \sqrt{10}$, $BC = \sqrt{18}$, $CD = \sqrt{10}$, $DA = \sqrt{18}$. Diagonals: $AC = 2$ and $BD = \sqrt{36+16} = \sqrt{52}$. Opposite sides are equal but the diagonals are not, so it is a parallelogram.

Answer: (i) a square (ii) no quadrilateral (iii) a parallelogram

Q7 Find the point on the x-axis that is equidistant from $(2,-5)$ and $(-2,9)$.

SOLUTION

Let the point be $(x,0)$. Then $(x-2)^2 + 25 = (x+2)^2 + 81$.

$x^2 - 4x + 29 = x^2 + 4x + 85$ gives $-8x = 56$, so $x = -7$.

Answer: $(-7,0)$

Q8 Find the values of y for which the distance between $P(2, -3)$ and $Q(10, y)$ is 10 units.

SOLUTION

$(10-2)^2 + (y+3)^2 = 100$ gives $(y+3)^2 = 36$, so $y + 3 = \pm 6$.

Answer: $y = 3$ **or** $y = -9$

Q9 $Q(0, 1)$ is equidistant from $P(5, -3)$ and $R(x, 6)$. Find the values of x, and the distances QR and PR.

SOLUTION

$QP^2 = 25 + 16 = 41$ and $QR^2 = x^2 + 25$. Setting them equal, $x^2 = 16$, so $x = \pm 4$.

In both cases $QR = \sqrt{41}$.

If $R = (4, 6)$: $PR = \sqrt{1 + 81} = \sqrt{82}$. If $R = (-4, 6)$: $PR = \sqrt{81 + 81} = 9\sqrt{2}$.

Answer: $x = \pm 4$, $QR = \sqrt{41}$, **and** $PR = \sqrt{82}$ **or** $9\sqrt{2}$

Q10 Find a relation between x and y such that (x, y) is equidistant from $(3, 6)$ and $(-3, 4)$.

SOLUTION

$(x-3)^2 + (y-6)^2 = (x+3)^2 + (y-4)^2$.

$-6x - 12y + 45 = 6x - 8y + 25$ gives $12x + 4y - 20 = 0$.

Answer: $3x + y - 5 = 0$

2 Exercise 7.2

Q1 Find the point that divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.

SOLUTION

Section formula: $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right) = \left(\frac{2 \times 4 + 3 \times (-1)}{5}, \frac{2 \times (-3) + 3 \times 7}{5}\right)$.

Answer: $(1, 3)$

Q2 Find the points of trisection of the segment joining $(4, -1)$ and $(-2, -3)$.

SOLUTION

Let $A = (4, -1)$ and $B = (-2, -3)$. The first point P divides AB in the ratio $1 : 2$: $P = \left(\frac{-2 + 8}{3}, \frac{-3 - 2}{3}\right) = \left(2, -\frac{5}{3}\right)$.

The second point Q divides AB in the ratio $2 : 1$: $Q = \left(\frac{-4 + 4}{3}, \frac{-6 - 1}{3}\right) = \left(0, -\frac{7}{3}\right)$.

Answer: $\left(2, -\frac{5}{3}\right)$ **and** $\left(0, -\frac{7}{3}\right)$

Q3 On a rectangular ground $ABCD$, lines are drawn 1 m apart, and 100 flower pots stand 1 m apart along AD. One student runs $\frac{1}{4}$ of AD along the 2nd line and posts a green flag. Another runs $\frac{1}{5}$ of AD along the 8th line and posts a red flag. Find the distance between the flags. Where should a blue flag go so that it is exactly halfway between them?

SOLUTION

Take A as the origin, AB as the x-axis and AD as the y-axis, with $AD = 100$ m.

Green flag: on the 2nd line at $\frac{1}{4} \times 100 = 25$ m, that is $(2, 25)$. Red flag: on the 8th line at $\frac{1}{5} \times 100 = 20$ m, that is $(8, 20)$.

Distance $= \sqrt{(8-2)^2 + (20-25)^2} = \sqrt{61}$ m.

Mid-point $= \left(\frac{2+8}{2}, \frac{25+20}{2}\right) = (5, 22.5)$.

Answer: $\sqrt{61}$ m apart. The blue flag goes on the 5th line, 22.5 m from AB.

Q4 In what ratio does $(-1, 6)$ divide the segment joining $(-3, 10)$ and $(6, -8)$?

SOLUTION

Let the ratio be $k : 1$. Then $\frac{6k-3}{k+1} = -1$ gives $7k = 2$, so $k = \frac{2}{7}$.

Check with y: $\frac{-8 \times 2 + 10 \times 7}{9} = 6$.

Answer: $2 : 7$

Q5 In what ratio does the x-axis divide the segment joining $A(1, -5)$ and $B(-4, 5)$? Find the point of division.

SOLUTION

Let the ratio be $k : 1$. The y-coordinate of the point is 0: $\frac{5k-5}{k+1} = 0$, so $k = 1$.

The point is the mid-point: $\left(\frac{1-4}{2}, 0\right) = \left(-\frac{3}{2}, 0\right)$.

Answer: $1 : 1$**, at** $\left(-\frac{3}{2}, 0\right)$

Q6 The points $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$, in that order, are the corners of a parallelogram. Find x and y.

SOLUTION

The diagonals of a parallelogram bisect each other, so the two diagonals have the same mid-point.

$\frac{1+x}{2} = \frac{4+3}{2}$ gives $x = 6$, and $\frac{2+6}{2} = \frac{y+5}{2}$ gives $y = 3$.

Answer: $x = 6, y = 3$

Q7 AB is a diameter of a circle with centre $(2, -3)$, and B is $(1, 4)$. Find A.

SOLUTION

The centre is the mid-point of AB. Let $A = (x, y)$: $\frac{x+1}{2} = 2$ gives $x = 3$, and $\frac{y+4}{2} = -3$ gives $y = -10$.

Answer: $A = (3, -10)$

Q8 A and B are $(-2, -2)$ and $(2, -4)$. Find the point P on the segment AB such that $AP = \frac{3}{7}AB$.

SOLUTION

$AP = \frac{3}{7}AB$ means $PB = \frac{4}{7}AB$, so P divides AB in the ratio $3 : 4$.

$P = \left(\frac{3 \times 2 + 4 \times (-2)}{7}, \frac{3 \times (-4) + 4 \times (-2)}{7}\right) = \left(-\frac{2}{7}, -\frac{20}{7}\right)$.

Answer: $P = \left(-\frac{2}{7}, -\frac{20}{7}\right)$

Q9 Find the points that divide the segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

SOLUTION

The middle point is the mid-point of AB: $(0, 5)$.

The other two are the mid-points of each half: $\left(\frac{-2+0}{2}, \frac{2+5}{2}\right) = \left(-1, \frac{7}{2}\right)$ and $\left(\frac{0+2}{2}, \frac{5+8}{2}\right) = \left(1, \frac{13}{2}\right)$.

Answer: $\left(-1, \frac{7}{2}\right)$, $(0, 5)$ **and** $\left(1, \frac{13}{2}\right)$

Q10 Find the area of the rhombus with vertices $(3, 0)$, $(4, 5)$, $(-1, 4)$ and $(-2, -1)$ taken in order.

SOLUTION

Call the vertices A, B, C, D. Diagonals: $AC = \sqrt{(-4)^2 + 4^2} = 4\sqrt{2}$ and $BD = \sqrt{(-6)^2 + (-6)^2} = 6\sqrt{2}$.

Area $= \frac{1}{2} \times AC \times BD = \frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} = 24$.

Answer: 24 square units

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